

DBA5101 – Analytics in Managerial Economics

Homework 2

This homework is worth 70 marks. Please submit your answers as a single PDF file through Canvas no later than **October 26, 2025, 23:59**. Late submissions will not be accepted.

Problem 1

An amusement park has two types of visitors: thrill-seekers and families. The demand functions for each group are as follows:

- **Thrill-Seekers (T):** $P_T = 70 - 2Q_T$
- **Families (F):** $P_F = 50 - Q_F$

The marginal cost (MC) of one family or a thrill-seeker customer is constant at SGD 10.

- (a) If the park can perform price discrimination between thrill-seekers and families, what prices should it charge each group to maximize profits?
- (b) If the park must charge the same price to all visitors, what is the optimal uniform price and total quantity sold to thrill-seekers and families?
- (c) Calculate the total profit in both scenarios (with and without price discrimination) and determine which strategy is more profitable. Explain briefly.

Problem 2

Suppose a movie theater can identify two distinct groups of customers, students and non-students. The demand by students Q_S and the demand by non-students Q_N are given by

- **Students (S):** $Q_S = 100 - 8P_S$
- **Non-students (N):** $Q_N = 100 - 4P_N$

The movie theater's cost of SGD 2 per customer is constant regardless of the number of customers served.

- (a) What price maximizes profits if the movie theater charges everyone the same price?
- (b) If the movie theater charges different prices for the two groups, then what prices will it charge to maximize its profits?
- (c) Calculate the profits and corresponding quantities for the two cases.
- (d) Compare the prices, quantities, and profits. What can you conclude?